Artifact 15 – Writing Linear Equations

Driving Question: When a relationship does not start at zero, how do you find its equation from words, a table, a graph, or two values?

Example 1

A storage unit rents for 65 dollars a month. A customer also pays a 20-dollar move-in fee on the first day. Write an equation for y, the total paid after x months.

Rate of change, m, and what it means

Total paid after 8 months

Initial value, b, and what it means

Proportional? Justify.

Equation

Example 2

Store B from your board: 30 dollars a week, plus a 55-dollar delivery fee on day one.

Weeks	Total paid
0	
4	
10	
26	

Rate of change, m, and initial value, b

Equation

Does $y = 30x$ fit store B? Use a row to decide.

Where store B's line meets the y-axis, and how it compares with store A's line

Example 3

A music school charges a registration fee plus the same price for each lesson. One family had paid 205 dollars in all after 5 lessons and 422 dollars in all after 12 lessons. Write an equation for y, the total paid after x lessons.

Rate of change, m. Show the work.

Equation

Check it with the value you did not use

Initial value, b. Show the work.

Notes to Forgetful Self

Given	Where m comes from	Where b comes from
words		
a table		
a graph		
two values		

A proportional relationship $y = kx$, written as $y = mx + b$: